MZUZU UNIVERSITY

Department of Mathematics

EMT 1101 - Precalculus

End of Semester Examinations

Monday, July 11, 2016 **Time:** $8:00 - 11:00$ **(3 hours)**

1. Check that the paper has **10 questions**.
2. Attempt all questions
3. Show all your working clearly
4. Use proper notation
5. Simplify your answers as far as possible
6. **No Calculators**

1. (a) Simplify $2\sqrt[3]{81} + 3\sqrt[3]{-24}$. (5)

   (b) Simplify $x(2x-2y)^{-1} + y(2y-2x)^{-1}$. (4)

2. (a) Simplify $|x-2| + |x-3|$ given that $x < 2$. (3)

   (b) Find all real solutions of the equation (4)

$$\frac{2y-5}{4y+1} = \frac{y-1}{2y+5}.$$

3. (a) Find the equation of a line that passes through the point $(1, 2)$ and is perpendicular to the line $x - 2y + 1 = 0$. (4)

   (b) Which point is farther from the origin, $(3, 9)$ or $(5, 8)$? (3)

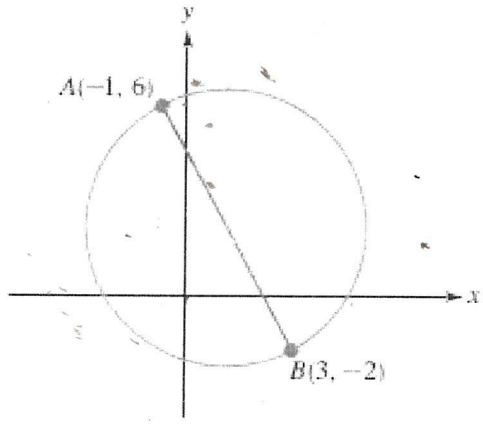
4. Test the equation $y^2 - y = 5x^2 + x$ for symmetry about:

   (a) the x−axis, (2)

   (b) the y−axis, (2)

   (c) the origin. (2)

5. The points $A(-1, 6)$ and $B(3, -2)$ are the endpoints of a diameter of a circle, as indicated in the following figure.



   (a) Find the equation of the circle. (6)

   (b) Find the y-intercepts of the circle. (2)

6. (a) Find a quadratic equation that has integer coefficients and the following roots: $\frac{1+\sqrt{5}}{3}$ and $\frac{1-\sqrt{5}}{3}$. (3)

   (b) Find all real solutions of each of the equation $1 + 14x^{-1} + 48x^{-2} = 0$. (4)

7. Solve the following inequality leaving your answer in interval notation: (10)

$$\frac{x^2-1}{x^2+8x+15} \geq 0.$$

8. (a) Let $h(x) = x^2 - 4x - c$. Find a nonzero value c such that $h(c) = c$. (3)

   (b) Use translations to sketch the graph of $y = -\frac{1}{x+2} + 3$. (6)

9. Suppose that in a certain biology lab experiment, the number of bacteria is related to the temperature T of the environment by the function

$$N(T) = -2T^2 + 240T - 5400.$$

Here, $N(T)$ represents the number of bacteria present when the temperature is T degrees Fahrenheit. Also, suppose that t hours after the experiment begins, the temperature is given by

$$T(t) = 10t + 40.$$

   (a) Find $N(T(t))$. (2)

   (b) How many bacteria are present when $t = 2$ hours? (2)

10. (a) Let $f(x) = \dfrac{1}{1-x}$:

      i. find the $f^{-1}(x)$, (3)

      ii. find that $f \circ f$. (3)

      iii. What do you observe about f^{-1} and $f \circ f$? (1)

   (b) Solve for x given that $f^{-1}(2x+3) = 5$ and $f(5) = 13$. (3)